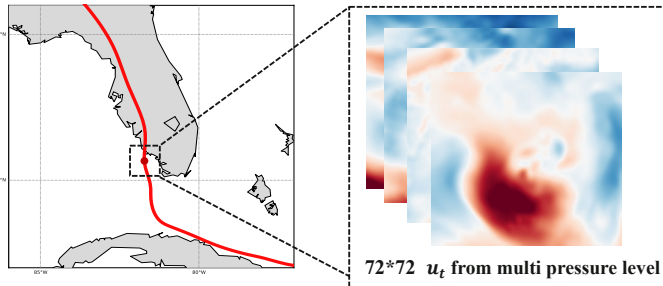


(A) FAST-ML

(1) Data Prepare



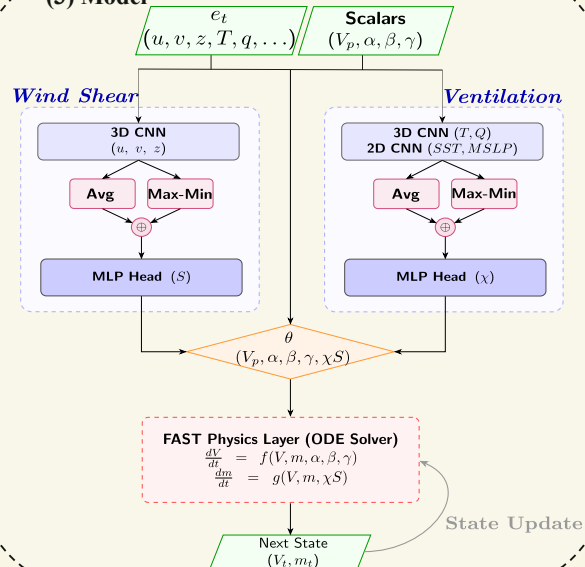
$$\text{IBTrACS Tracks} \rightarrow e_t = [u_t, v_t, z_t, T_t, q_t, SST_t, MSLP_t]$$

(2) Scalars Compute

$$V_p^2 = S_w^2 \frac{C_k T_s}{C_d T_o} (CAPE^* - CAPE)$$

$$\alpha = 1 - 0.87e^{-z} \quad \beta = 1 - \epsilon - \kappa \quad \gamma = \epsilon + \alpha \kappa$$

(3) Model



(B) Forecast Pipeline

